NOT MEASUREMENT SENSITIVE

MIL-I-85071A(AS)
31 July 1992

SUPERSEDING
MIL-I-85071(AS)
16 March 1979

MILITARY SPECIFICATION

INVERTERS, AIRCRAFT, DC TO AC, GENERAL SPECIFICATION FOR

This specification is approved for use by the Naval Air Systems Command, Department of the Navy, and is available for use by all Departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for self-cooled inverters for use in converting electrical power in aircraft from nominal 28 Volts DC (see 6.4.2) power to three-phase or single-phase AC (see 6.4.1) power at 400 Hertz. Coverage includes inverters capable of supplying regulated output with the low input voltages typically encountered during engine starting.

1.2 Classification. Inverters shall be of the following classes, as specified (see 3.7.2.1 and 6.2):

Class A. Class A inverters shall be designed to meet the temperature-altitude requirements of MS33543, Curve II, except that no operation shall be required below -55°C or above 50,000 feet above sea level.

Class B. Class B inverters shall be designed to meet the temperature-altitude requirements of MS33543, Curve I, except that no operation shall be required below -55°C, above +85°C, or above 65,000 feet above sea level.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304, by using the self-addressed Standardization Document Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

FSC 6130

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.3).

SPECIFICATIONS

Military

MIL-P-116	Preservation-Packaging Methods of
MIL-W-5088	Wiring, Aerospace Vehicle
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-S-7742	Screw Threads, Standard, Optimum Selected Series; General Specification for
MIL-T-7928	Terminal, Lug and Splice, Crimp Style Copper
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-P-15024	Plate Tags and Bands for Identification of Equipment
MIL-S-19500	Semiconductor Device, General Specification for
MIL-A-21180	Aluminum Alloy Castings, High Strength
MIL-C-39006/22	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR79
MIL-C-39006/25	Capacitor, Fixed, Electrolytic (Nonsolid Electrolyte), Tantalum, (Polarized, Sintered Slug), 85°C (Voltage Derated to 125°C), Established Reliability, Style CLR81
MIL-R-39015	Resistors, Variable, Wirewound (Lead Screw Actuated) Established Reliability, General Specification for
MIL-I-46058	Insulation Compound, Electrical (For Coating Printed Circuit Assemblies
MIL-E-81910	Electric Power and Control Equipment, Aircraft, Testing, General Specification for
MIL-I-85071/1	Inverter, Aircraft, DC to AC, Static, 100VA 3-Phase and 100VA 1-Phase, Multipurpose, Class B

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

MIL-I-85071/2	Inverter, Aircraft, DC to AC, Static, 250VA 3-Phase and 250 VA 1-Phase, Multipurpose, Class B
MIL-I-85071/3	Inverter, Aircraft, DC to AC, Static, 750VA 3-Phase, 1000VA 1-Phase, and 1000VA 3-Phase, Multipurpose, Class B

STANDARDS

Federal

FED-STD-595	Colors

Military

MIL-STD-129	Marking for Shipment and Storage
MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-198	Capacitors, Selection and Use of
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-481	Configuration Control-Engineering Changes (Short Form), Deviations and Waivers
MIL-STD-785	Reliability Program for Systems and Equipment Development and Production
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-831	Test Report, Preparation of
MIL-STD-970	Standards and Specifications, Order of Preference for the Selection of
MIL-STD-1562	Lists of Standard Microcircuits
MIL-STD-1629	Procedures for Performing a Failure Mode Effects and Criticality Analysis
MIL-STD-2000	Standard Requirements for Soldered Electrical and Electronic Assemblies
MIL-STD-2073-1	DOD Materiel Procedures for Development and Application of Packaging Requirements
MIL-STD-2073-2	Packaging Requirement Codes
MIL-STD-2175	Castings, Classification and Inspection of
MS3376	Nuts, Self-locking, Cres, 12 Point Captive Washer
MS33543	Criteria-Temperature and Altitude Range Self Cooled Electric Equipment

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

MS90415 Nut, Self Locking, Steel, 160 KSI, 450 DEG. F, 12 Point Captive Washer

MILITARY HANDBOOKS

MIL-HDBK-217 Reliability of Electronic Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (Attn: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

NAVAL AIR SYSTEMS COMMAND DOCUMENTS

AS-4613 Application and Derating Requirements for Electrical Components

(Copies are available from the Naval Air Systems Command, (Attn: AIR-5112), Jefferson Plaza 2, Washington, DC 20361-5110.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications (see 6.4.3)), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specifi exemption has been obtained.

REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheets. In the event of any conflict between requirements of this specification and the specification sheet, the latter shall govern. (If a specific requirement specified herein is not required for an inverter, it shall be so indicated on the specification sheet; for example "Shock - Not applicable.").

3.2 Classification of requirements. The requirements for inverters are classified herein as follows:

Requirement	Paragraph
Qualification	3.3
Materials	3.4
Selection of parts and materials	3.5
Design and construction	3.6
Performance characteristics	3.7
Reliability	3.8
Markings	3.9
Workmanship	3.10

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

3.3 Qualification. Inverters furnished under this specification shall be products which are qualified for listing on the applicable Qualified Products List at the time set for opening of bids (see 4.4 and 6.3). Each unit delivered shall fully meet all the requirements of this specification and the applicable specification sheet, and shall be capable of successfully completing the tests required by this specification for quality conformance inspection and for inspections listed for any one of the qualification samples. The Naval Air Warfare Center Aircraft Division (SY60), Patuxent River, Maryland 20670-5304 is the qualifying activity for equipment submitted under this specification. Any design changes to the inverter shall be handled in accordance with MIL-STD-481, with copies of the documentation provided to the qualifying activity and, if applicable, the procuring activity.

3.4 Materials. Materials used in the manufacture of inverters shall be of high quality and suitable for the intended purpose.

3.4.1 Metals. Metals shall be of a corrosion resistant type or suitably processed to resist corrosion.

3.4.1 Dissimilar metals. Unless otherwise suitably protected, dissimilar metals such as brass, copper, or steel shall not be used in intimate contact with aluminum or its alloys. When protection is used, it shall be of such type that a low impedance path is offered to radio frequency currents. Dissimilar metals are defined as two metal specimens that generate an electric current in a conductive solution when in contact or otherwise electrically connected.

3.4.1.2 Magnesium alloy parts. Neither magnesium nor magnesium alloy parts shall be used.

3.4.1.3 Finishes and protective coatings.

3.4.1.3.1 Aluminum alloy parts. Aluminum alloy parts shall be anodically treated in accordance with MIL-A-8625 or chemically treated in accordance with MIL-C-5541. To provide adequate bonding, the aluminum oxide film formed by this treatment shall be removed from the areas under screws, nuts, etc. used for assembly or mounting purposes.

3.4.1.3.2 Steel parts. Steel parts shall be plated, if necessary, so that the inverter meets all of the specified electrical and environmental requirements.

3.4.1.3.3 Color. The inverter shall be finished in 37038 black in accordance with FED-STD-595.

3.4.1.4 Castings. Castings shall be in accordance with MIL-STD-2175 and shall be Class 3, Grade C or better. Aluminum alloy castings shall be in accordance with MIL-A-21180. Defects not materially affecting the suitability of the castings may be repaired at the foundry or during machining by peening, impregnating, welding, or other methods acceptable to the procuring activity. Inspection of repaired castings shall be governed by quality control techniques subject to the approval of the procuring activity.

3.4.2 Non-metallic materials. Non-metallic materials used shall be non-toxic, fungus inert, flame resistant, and shall not support combustion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

3.5 Selection of parts and materials. Specifications and standards for all materials and parts, and Government certification and approval of processes and equipment, which are not specifically designated herein and which are necessary for the design and construction of inverters complying with the requirements of this specification, shall be selected in accordance with MIL-STD-970.

3.5.1 Standard parts. Military standard parts shall be used wherever they are suitable for the purpose, and shall be identified on the drawing by their military part numbers. Any standard part, of the same part number, shall be readily interchangeable as applied within the inverter.

3.5.2 Non-standard parts. The use of non-standard parts is permitted in the event that there is no suitable standard part available on the date of invitation for bids, however, such use shall be approved by the qualifying activity. If a non-standard part is approved by the qualifying activity, subsequent approval is not necessary. Request for approval for the use on non-standard parts shall be submitted in writing and shall contain the following information:

a. Designation of equipment and assembly wherein part is used.

b. Reference designation of part (circuit, symbol, part number, as applicable).

c. Name and complete detailed technical description of part.

d. Part manufacturer's name and address.

e. Part manufacturer's drawing and part number.

f. Reason for not using standard part including, when applicable, a comparison of characteristics of the non-standard part with the standard part which most nearly possesses the required characteristics.

g. Other information required herein for specific types of parts.

3.5.3 Commercial (hardware) parts. Commercial utility hardware items: screws, bolts, nuts, and cotter pins may be used without approval provided they possess suitable properties and are replaceable by MS or AN parts without alteration, and provided the corresponding MS or AN part numbers are referenced in the parts list and on the manufacturer's drawings.

3.5.4 Semiconductor devices. Military standard semiconductor devices shall be chosen, applied, and reported in accordance with Requirement 30 of MIL-STD-454 and shall be selected from those meeting MIL-S-19500 and listed in QPL-19500. The reliability level of devices selected shall be in the following order of preference beginning with the most preferred: JANTXV, JANTX, JAN. Requests for approval to use non-standard semiconductors shall be accompanied by substantiating reliability data or proposed testing to assure adequate device reliability for the application.

3.5.5 Microelectronic devices. Microelectronic devices shall be chosen, applied, and reported in accordance with Requirement 64 of MIL-STD-454 and shall be listed in MIL-STD-1562. Requests for approval to use devices which do not comply with this

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

requirement shall be submitted as non-standard parts, and shall be accompanied by substantiating reliability data or proposed testing to assure adequate device reliability for the application.

3.5.6 Capacitors. Capacitors shall be selected and applied in accordance with MIL-STD-198. The use of wet slug tantalum capacitors, except for tantalum cased units in accordance with MIL-C-39006/22 and MIL-C-39006/25, requires the approval of the qualifying activity and, if applicable, the procuring activity. Silver cased tantalum capacitors shall not be used.

3.5.7 Electron tubes. Electron tubes of either the vacuum or gas filled type shall not be used.

3.5.8 Potentiometers and adjustable resistors. Only sealed units conforming to MIL-R-39015 shall be used unless approved as a non-standard part.

3.6 Design and construction. The inverter shall be a single package, static, and solid-state unit having no moving parts, except as noted in 3.6.2.7. Dynamic protective devices (fuses, circuit breakers, and thermally actuated switches) shall not be used. Any changes to the design of the inverter shall be handled as specified in 3.3.

3.6.1 Electrical.

3.6.1.1 Component derating. All inverter electric/electronic components shall be derated as shown in Table I based on the most severe inverter environmental and operational conditions. Stress levels in excess of the above shall be submitted to the qualifying activity for written approval along with the rationale for the use of that stress level in the application.

3.6.1.2 Electric connections.

3.6.1.2.1 External electric connections. The equipment shall be so designed that external connections and wiring can be made in accordance with the wiring installation, ground return, and connectors requirements of MIL-W-5088. The equipment structure shall not be used as a current path except for radio noise shielding.

3.6.1.2.1.1 Terminal block design. Where terminals are used for the external connection to inverters, they shall be of the stud type, and shall be designed so that the current is conducted by means of surface-to-surface contact, and not through the stud threads. All studs shall be steel, corrosion resistant, C-34 Rockwell minimum hardness, and shall be of sufficient length to accommodate to MIL-T-7928 lugs. Nuts used on terminals shall conform to either MS3376 or MS90415. There shall be no dielectric material in the compression build-up. The terminal block shall be so designed that it can be removed and replaced on the inverter without the necessity of rebrazing or soldering. Barriers affording a positive separation of leads and terminals shall be provided on the terminal block. Durable, reusable, non-conductive terminal covers shall be provided. Terminal designations shall be durably, legibly, and prominently marked on the terminal block.

3.6.1.2.1.2 Connectors. When alternate connectors are specified, the inverter design shall be such that it may be loaded on either receptacle, but not simultaneously. This shall be accomplished by a simple, easily accessible means to

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

electrically disconnect the unused receptacle, or by providing a suitable protective cover which will automatically cover the unused receptacle. The cover, if used, shall be designed so that it cannot be separated from the unit.

3.6.1.2.1.3 Output voltage jacks. One set of placarded pin jacks shall be provided on the inverter located as shown on the applicable specification sheet between which the output voltage can be measured. The pin jacks shall accept a probe pin of .078 ± .002 inches in diameter. A set shall consist of two jacks for single-phase units and four jacks for three-phase units. Jack designations shall be durably, legibly, and prominently marked on the unit.

3.6.1.2.2 Internal electric connections. Internal connections shall be made in accordance with MIL-STD-454, Requirements 1, 10, 17, 19, 20, and 69. General purpose soldering shall be performed in accordance with MIL-STD-2000 Task A. For applications which include active surface mounted microelectronic devices, soldering shall be performed in accordance with MIL-STD-2000 Task F. For three-phase inverters, the design shall be such that no internal reconnections will be required to change from wye to delta.

3.6.1.2.3 Adjustments. The inverter shall not have any external provisions for voltage or frequency adjustment, nor shall any internal adjustments be necessary at the time of installation. All adjustments shall be made at the time of manufacture or overhaul.

3.6.2 Mechanical.

3.6.2.1 Maintainability. The inverter shall be designed to facilitate assembly disassembly, location of trouble sources, and maintenance without aid of special tools. All functional parts shall be readily identifiable, accessible, and removable for replacement. When printed circuit boards are utilized, only clear transparent conformal coating conforming to MIL-I-46058 type UR or XY shall be used in order to permit identification, location, removal, and replacement of functional parts. Functional parts are defined as individual component parts such as resistors, capacitors, semiconductor devices, inductive devices, or replaceable functional assemblies such as active or passive filters, voltage dividers, dual transistors or diodes.

3.6.2.2 Failure mode and effects analysis (FMEA). A failure mode and effects analysis shall be performed in accordance with the requirements of MIL-STD-1629 Task 101. Inverter protective functions shall be independent of the control functions, and no single fault or failure shall result in inverter operation outside the protective limits specified herein.

3.6.2.3 Shock mounts. Shock mounts (or vibration isolators) shall not be used.

3.6.2.4 Interchangeability. Provisions for interchangeability are required to the maximum extent possible. Design tolerances shall permit parts, subassemblies, and assemblies to be used in the inverter without regard to the source of supply or manufacturer. Parts, subassemblies, and assemblies having the full range of dimensions and characteristics permitted by the governing specification shall be usable as replacement items without selection and without departure from the specified performance requirements of the inverter. When permission is granted to use a non-standard part or material because the existing standard part or material is not

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

available, the inverter shall be so designed that the non-standard part or material and the standard part or material are interchangeable. When the specification for the part or material contains substitutability or supersession information, the design shall permit the substitute or superseding parts or materials to be used interchangeably.

3.6.2.5 Screw threads. Screw threads shall conform to the requirements of MIL-S-7742.

3.6.2.6 Safety wiring and staking. Accidental loosening of screws, screw parts, and other connections shall be prevented by safety wiring (0.032 inch minimum OD) where practicable, staking, self-locking nuts, or other approved methods. Washers and cotter pins, where used, shall be assembled in a manner which prevents rotation of washers and movement of cotter pins under conditions of vibration.

3.6.2.7 Cooling. The inverter shall be self-cooled. No conduction cooling to the inverter environment shall be allowed. Safe operating equilibrium temperature shall be maintained for the inverter over the specified range of operating conditions, and the rated temperature of individual inverter components shall not be exceeded. The use of an internally mounted, integral cooling fan is permitted for inverters with ratings of 750 VA and greater. The fan shall use a motor having ball bearings of quality level ABEC-3 or higher, and the fan shall be rated for operation at 100°C ambient or higher.

3.6.2.8 Ventilation openings. All ventilation or other openings shall be designed to prevent passage of foreign objects, and shall be of such size to prevent passage of spherical objects of one-fourth inch diameter or larger.

3.6.2.9 Operating position. The inverter shall operate in any position.

3.6.2.10 Size and weight. The dimensions and weight shall be as specified in the applicable specification sheet. The weight and dimensions shown on the applicable specification sheet shall include the weights and dimensions of all auxiliary apparatus necessary to make the inverter conform to this specification and the applicable specification sheet. Where only maximum and minimum dimensions are shown, the part need not have the shape shown. But, the part, including all protrusions, shall be contained within the outline shown.

3.7 Performance characteristics. The inverter shall deliver power in accordance with the requirements specified herein and the applicable specification sheet. For three-phase units, the limits apply to each phase.

3.7.1 Electrical.

3.7.1.1 Input power. The inverter shall meet the requirements of this specification when supplied with power as defined herein from an aircraft electrical system. The nominal input voltage is 28 Volts DC.

a. Normal input voltage (all inverters) - The normal input voltage range is indicated on the applicable specification sheet and is defined as the range encountered when all aircraft engines are operating and the aircraft electrical system is fully operational.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
Check the source to verify that this is the current version before use.

b. Abnormal input voltage (all inverters) - The abnormal input voltage range is indicated on the applicable specification sheet and is defined as the range encountered during operation from aircraft batteries or other emergency power sources.

c. Low input voltage - The low input voltage range, when applicable, is indicated on the applicable specification sheet and is defined as the range encountered during operation from aircraft batteries when those batteries are being used for engine starting. Performance of inverters at low input voltage conditions shall be as specified in 3.7.1.2 and on the applicable specification sheet.

3.7.1.2 Output voltage regulation. The inverter shall maintain the following steady state output voltages under all specified conditions (see 4.7.2.2 and 4.7.2.4):

a. Normal input voltage (see 3.7.1.1a)

(1) Single-phase and balanced three-phase loads: 112.5 - 117.5 Volts RMS

(2) One-third unbalanced three-phase loads: 112.5 - 118.0 Volts RMS

(3) Two-thirds unbalanced three-phase loads: 112.0 - 119.0 Volts RMS

b. Abnormal input voltage (see 3.7.1.1b) for single-phase and balanced three-phase loads: 108.0 - 117.5 Volts RMS

c. Low input voltage (see 3.7.1.1c) for single-phase and balanced three-phas loads: 45.0 - 117.5 Volts RMS

d. Low input voltage (inverters required to supply regulated output at low input voltages)(see 3.7.1.1c) for single-phase and balanced three-phase loads: 108.0 - 117.5 Volts RMS

3.7.1.3 Output frequency regulation. The inverter shall maintain a steady state frequency of 400 ± 7 Hertz under all specified input voltages (see 3.7.1.1), load, temperature, and altitude conditions specified (see 4.7.2.2).

3.7.1.4 Efficiency. The efficiency of the inverter shall be as follows under the conditions specified (see 4.7.2.3):

RATING (VA)	MINIMUM EFFICIENCY (Percent)
100 and below	60
101 through 1000	65
over 1000	70

3.7.1.5 Three-phase output.

3.7.1.5.1 Phase rotation. The phase rotation shall be in the order of A-B-C (see 4.7.2.4.2).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

3.7.1.5.2 Output voltage unbalance. For all balanced and one-third unbalanced load conditions, the maximum output voltage unbalance shall be 3 Volts RMS. For two-thirds unbalanced load conditions, the maximum output voltage unbalance shall be 4 Volts RMS. Individual phase voltages (see 6.4.4) shall be within the limits specified in 3.7.1.2.

3.7.1.5.3 Phase displacement. The displacement between adjacent phases shall be 120 ± 4° for both balanced and unbalanced load conditions (see 4.7.2.4.3).

3.7.1.6 Recovery time. When rated full load (see 6.4.3) is applied and removed, the output voltage shall remain within the limits of Figure 1 (see 4.7.2.5).

3.7.1.7 Input power application. The output voltage during input power application shall be within the limits of the applicable specification sheet within two seconds after application of input voltage. During the two second period, output voltage transients shall be within the limits of Figure 1. For determining compliance of voltage transients to Figure 1, the onset of the transient shall be considered as the time when the output voltage first reaches a value equal to minimum steady state limits. The input power application current shall not exceed 1000 percent of full load current as computed from the minimum allowable efficiency specified (see 3.7.1.4). The current shall return to and stay within 150 percent of full load value within 0.1 seconds after first application of input voltage (see 4.7.2.6).

3.7.1.8 Overload and short circuit capacity. Over the normal input voltage range (see 3.7.1.1a) the inverter shall be capable of operating without damage into any passive load impedance (including a short circuit) for five seconds. For three-phase units this shall apply to each phase separately, any two phases simultaneously, or all phases simultaneously (see 4.7.2.7).

3.7.1.8.1 Overload. Over the normal input voltage range, inverters rated 1000 VA or less shall, without damage, maintain the minimum output voltage specified by the applicable specification sheet while delivering 200 percent rated output current for five seconds. Inverters rated above 1000 VA shall maintain the minimum output voltage specified by the applicable specification sheet while delivering 150 percent rated output current for 15 seconds. For three-phase units, this shall apply to each phase separately, any two phases simultaneously, or all three phases simultaneously. During these conditions, no phase voltage shall exceed 117.5 Volts RMS.

3.7.1.8.2 Short circuit. Over the normal input voltage range, inverters rated 1000 VA or less shall be capable of delivering, without damage, at least 250 percent rated load current for five seconds. Inverters rated above 1000 VA shall be capable of delivering, without damage, at least 200 percent rated load current for five seconds. For three-phase units, short circuit requirements shall apply to one phase, two phase, and three phase line-to-line and line-to-neutral short circuits. During these conditions, no phase voltage shall exceed 117.5 Volts RMS.

3.7.1.8.3 Protection. After delivering current into any overload or short circuit condition for at least five or 15 seconds (as applicable), the inverter shall have automatic electronic overload and short circuit protection and shall then operate continuously for at least 15 minutes, without damage, while connected to the overload or short circuit. The output voltage and/or current may drop to any level after the initial five or 15 seconds (as applicable). Normal output voltage shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

automatically restored within two seconds after the removal of the overload or short circuit. For three-phase units this requirement shall apply to each phase separately, any two phases simultaneously, or all phases simultaneously.

3.7.1.9 Electromagnetic compatibility. When tested as specified in 4.7.2.8, the conducted and radiated electromagnetic interference shall be within the limits specified in MIL-E-81910.

3.7.1.10 Output modulation. (See 4.7.2.9)

3.7.1.10.1 Voltage. The output voltage modulation shall not exceed 1.0 percent under normal input voltage conditions, and 1.5 percent under abnormal and low input voltage conditions (see 3.7.1.1).

3.7.1.10.2 Frequency. The output frequency modulation shall not exceed 0.07 percent, nor shall the deviation rate exceed 200 Hertz per second.

3.7.1.11 Waveform. (See 4.7.2.10)

3.7.1.11.1 Linear loads (full wave sinusoidal output current). The output voltage crest factor shall not exceed the limits of 1.414 ± .1 over the normal input voltage range. The crest factor shall not exceed the limits of 1.414 ± .3 over the abnormal input voltage range. The total harmonic and extraneous frequency content of the waveform shall not exceed 5 percent of the fundamental, nor shall any individual harmonic or extraneous frequency exceed 3 percent of the fundamental, over the normal input voltage range (see 3.7.1.1). At temperature and altitude conditions outside c those specified in 4.7.1.1 and 4.7.1.2 the 3 percent limit shall not apply.

3.7.1.11.2 Nonlinear loads. During the tests specified in 4.7.2.10.1.2, the total harmonic and extraneous frequency content of the waveform shall not exceed 8 percent of the fundamental.

3.7.1.12 Input ripple current. Peak-to-mean input ripple currents shall not exceed the values specified below, when the inverter is operated from a low impedance, low ripple source (see 4.7.2.11).

INVERTER RATING - VA	PEAK-TO-MEAN INPUT RIPPLE CURRENT MAX-AMP
Under 30	0.5
30-90	1
91-200	2
201-500	5
501-1000	10
1001-2500	20
over 2500	25

3.7.1.13 Input power susceptibility. The inverter shall not be damaged by input surge and spike voltages defined in 4.7.2.12. There shall be no evidence of flashover or insulation breakdown as a result of the tests specified in 4.7.2.12 and the output transients generated during the test shall not exceed the limits defined in Figure 1

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
Check the source to verify that this is the current version before use.

During the test specified in 4.7.2.12.3, the output voltage, frequency, output modulation, and waveform of the inverter shall be within the limits specified in 3.7.1.2a, 3.7.1.3, 3.7.1.10, and 3.7.1.11.

3.7.1.14 Extreme input voltage. The inverter shall be capable of carrying rated load when operated continuously at the minimum abnormal input voltage. The inverter shall not be damaged when energized continuously at input voltages from zero to 32 Volts under any rated load condition. The inverter shall maintain output voltage between the maximum normal input, and 32 Volts input, however, no other requirements shall apply except output voltage shall not exceed 117.5 Volts RMS under these conditions (see 4.7.2.13).

3.7.1.15 Acoustical noise. When limits are specified by the applicable specification sheet and when tested as specified in 4.7.2.14, the maximum sound level measured at any point shall not exceed the specified limits.

3.7.2 Environmental.

3.7.2.1 Temperature-altitude. The inverter shall be capable of operation without degradation of performance throughout the temperature-altitude range specified below for the class specified in the applicable specification sheet.

3.7.2.1.1 Class A. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range of MS33543, Curve II, except that no operation shall be required below -55°C or above 50,000 feet above sea level.

3.7.2.1.2 Class B. The inverter shall be capable of operation without degradation in specified performance throughout the temperature-altitude range of MS33543, Curve I, except that no operation shall be required below -55°C, above +85°C, or above 65,000 feet above sea level.

3.7.2.2 Acceleration. The output voltage and frequency of the inverter shall remain within the limits of 3.7.1.2 and 3.7.1.3 while operating during the acceleration test (see 4.7.2.15). There shall be no evidence of physical damage, and the voltage and frequency regulation after the test shall be within the limits specified in 3.7.1.2 and 3.7.1.3.

3.7.2.3 Humidity. As a result of the humidity test there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.16).

3.7.2.4 Shock.

3.7.2.4.1 Basic design. The output voltage shall not exceed the limits of Figure 1 while operating during the basic design shock tests (see 4.7.2.17). There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:49Z
Check the source to verify that this is the current version before use.

3.7.2.4.2 Crash safety. The inverter shall comply with the equipment crash safety mechanical shock requirements of MIL-E-81910.

3.7.2.5 Vibration. The output voltage modulation of the inverter measured during the test shall not exceed two percent (see 4.7.2.18). The output voltage and frequency measured during the test shall remain within the limits specified in 3.7.1.2 and 3.7.1.3. There shall be no evidence of mechanical damage, loosening of parts or hardware, or physical distortion of the enclosure. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3.

3.7.2.6 Fungus. As a result of the fungus test the inverter shall show no evidence of fungus growth on any surface. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.19).

3.7.2.7 Salt fog. As a result of the salt fog test there shall be no evidence of corrosion, peeling, pitting, blistering, or cracking of the finish. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.20).

3.7.2.8 Dust. The inverter shall be capable of operating within the specified limits during the sand and dust test. At the conclusion of the test, the voltage and frequency regulation shall be within the limits specified in 3.7.1.2 and 3.7.1.3 (see 4.7.2.21).

3.7.2.9 Explosive atmosphere. The inverter shall not cause ignition when tes as specified in 4.7.2.22.

3.8 Reliability. Reliability shall be considered in each phase of the inverter design process. Techniques used to determine and evaluate the detailed design shall include a stress analysis with associated predictions and allocations in conformance with MIL-HDBK-217; sneak circuit and worst case analysis on all inverter control and protective circuits in accordance with MIL-STD-785, Tasks 205 and 206; and parts derating in accordance with the requirements of AS-4613 for class B equipment. Manufacturing screening of each inverter shall be provided by the random vibration and temperature cycling of the burn-in test. Inverters shall have a minimum observed Mean Flight Hours Between Failure (MFHBF) of 2,000 hours when operated under specified conditions.

3.8.1 Reliability development. The reliability of the inverter shall be demonstrated by a test, analyze, and fix (TAAF) test program (see 4.7.3).

3.9 Markings.

3.9.1 Terminals. Markings of external electrical connections shall be as specified on the applicable specification sheet.

3.9.2 Identification of product. Equipment, assemblies, and parts shall be marked in accordance with MIL-STD-130.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

3.9.3 Nameplate. A nameplate conforming to MIL-P-15024, Type A, B, C, G, or H, with the following markings, in addition to the requirements of MIL-STD-130, shall be securely attached to the inverter. The nameplate shall contain the following information:

Inverter	_____	VA		
Input:	_____	Volts DC	_____	Amperes DC
Output:	_____	Volts AC	_____	Amperes AC
	400	Hz	_____	Phase

Military Part No.
National Stock No.
Manufacturer's Part No.
Manufacturer's Serial No.
Date of manufacture
Contract or Order No.
Manufacturer's name or trademark
Inspector's stamp
US

3.9.4 Airflow. When an integral cooling fan is used, direction of airflow shall be plainly and permanently marked on the outside of the housing. When the direction of the airflow is marked on the cover, provision shall be made to assure that the cover cannot be installed to indicate a false airflow direction.

3.10 Workmanship. The inverter shall be manufactured and assembled in accordance with MIL-STD-454, Requirement 9.

3.10.1 Burn-in. Each inverter shall be subjected to a burn-in test as specified in 4.7.5 in order to expose and eliminate workmanship and component quality problems.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to the prescribed requirements.

4.1.1 Responsibility for compliance. All inverters shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all the requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

4.2 Classification of inspection. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (see 4.4).

b. Qualification retention inspection (see 4.5).

c. Quality conformance inspection (see 4.6).

4.3 Inspection conditions. Unless otherwise specified, all inspections shall be performed in accordance with the test conditions specified in 4.7.1.

4.4 Qualification inspection.

4.4.1 Qualification tests. Qualification tests shall be conducted at a facility satisfactory to the activity responsible for qualification (see 6.3). Qualification tests shall consist of the examination, inspections, and tests specified in Table II, in the order shown or as otherwise approved by the qualifying activity.

4.4.2 Qualification samples. The qualification test sample shall consist of at least four (4) units of each part number for which qualification is sought. The sample units shall be produced, to the maximum extent practicable, with equipments and procedures normally used in production, and shall be identified as specified (see 3.9). The samples shall have successfully completed the quality conformance inspections shown in Table III.

4.4.3 Data submission.

4.4.3.1 Drawings. Together with the test samples, the manufacturer shall submit two (2) reproducible copies of the outline and detail assembly drawings, schematics, and parts lists. Upon completion of qualification testing and prior to qualification approval, the manufacturer shall again provide two (2) reproducible copies of the outline and detail assembly drawings, schematics, and parts lists. Any changes in the reproducible copies from those submitted with the qualification samples shall be indicated in detail.

4.4.3.2 Certification. Together with the test samples, the manufacturer shall submit a materials inspection report consisting of a certification, supported by verifying data, stating that the materials and finishes specified in 3.4 and used in fabricating the inverters are in accordance with the applicable referenced specification or requirements prior to such fabrication. Also, the manufacturer shall submit documentation certifying that the inverter meets the explosive atmosphere requirements specified in 3.7.2.9.

4.4.3.3 Component stress level. Together with the test samples, the manufacturer shall submit a report listing all inverter electric/electronic components, their maximum rated stress, their applied stress within the inverter, and the ratio of applied to rated stress. An analysis of electric/electronic components shall be performed in accordance with MIL-HDBK-217 to demonstrate conformance to paragraph 3.8 of this specification. Data shall be based upon the most severe inverter environmental and operational conditions specified herein and the applicable specification sheet.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

4.4.4 Disposition of qualification samples and test report. Upon completion of the qualification tests, the test samples shall be identified as required and forwarded to the qualifying activity (see 3.3 or 6.3). The test samples shall be accompanied by three (3) copies of the qualification test report, prepared in accordance with MIL-STD-831.

4.4.5 Retention of qualification. To retain qualification, the manufacturer shall comply with the requirements of 4.5. In the event that no production occurred during a twelve (12) month period, a report shall be submitted to the qualifying activity certifying that the company still has the capabilities and facilities necessary to produce the item. If during two (2) consecutive twelve (12) month periods there has been no production, the manufacturer may be required, at the discretion of the qualifying activity, to submit a representative product sample for testing in accordance with the qualification inspection requirements.

4.5 Qualification retention inspections. Each inverter of the same part number delivered as a qualified item under this specification shall be numbered sequentially in essentially the order that it is submitted for Government acceptance, regardless of the contract under which it is delivered or by whom it is purchased. To meet the delivery requirements for the first 50 or less inverters, the contract delivery requirement plus one additional inverter must successfully complete the required quality conformance inspections. The contract delivery of required inverters are to be shipped as directed by the applicable contract(s). The additional inverter (retention sample), selected at random by the Government Inspector, is to be forwarded to the Naval Air Warfare Center Aircraft Division, Electrical Systems Department (SY60), Patuxent River, MD 20670-5304 (Reference: 101). To meet the delivery requirements for each succeeding 150 inverters, 151 inverters must successfully complete the required quality conformance inspection, 150 of which are to be shipped as directed by the applicable contract(s). The additional inverter (retention sample) shall be randomly selected and forwarded as above. The span of serial numbers and the contract number of the production units from which each retention sample was selected shall be recorded, and this information included with the retention sample. Each retention sample shall successfully complete the quality conformance inspection shown in Table III, and the test data and test date shall also be included with the inverter (retention sample). The samples shall be shipped no later than the final unit in the production group from which they are selected. The units forwarded to the qualifying activity will be inspected and tested for conformance to this specification and the applicable detail specification. Tests will be conducted for each inverter (retention sample) according to any column from Table II at the discretion of the qualifying activity. All retention samples become Government property when accepted by the Government Inspector for shipment to the qualifying activity. The cost for each inverter (retention sample) shall be borne by the manufacturer. These requirements shall apply to all inverters procured in accordance with this specification whether procured directly by the Government or by private industry procuring for end item use by the Government.

4.5.1 Qualification retention samples. For each quotation and based on all production contracts obtained, it will be the manufacturer's responsibility to inform the procuring activity of the exact number of qualification retention samples required for forwarding to the qualifying activity. The number of qualification retention samples required should be taken into account when the contract bid is prepared. If the proper number of retention samples is not itemized in the contract, then the manufacturer will, without additional cost to the Government, forward to the qualifying

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

activity whatever number of retention samples is determined by the cognizant Government Inspector to be required by paragraph 4.5 of this specification.

4.5.2 Rejection and retest of qualification and qualification retention samples. Samples which have been rejected or returned to the manufacturer for any reason during qualification or qualification retention tests, may be reworked or have parts replaced to correct defects. The manufacturer shall bear the cost of any repair or redesign work necessitated by a failure to meet the requirements. Before resubmitting the samples, full particulars concerning the rejection and the corrective action taken by the manufacturer must be submitted in writing by the manufacturer to the qualifying activity. Tests shall not be resumed until such a report is received. After the second attempt to meet one of the requirements of this specification or the applicable specification sheet, the manufacturer shall bear the cost of any additional retesting for that requirement. Where qualification tests are conducted under the auspices of the manufacturer, the qualifying activity will be advised upon failure of a qualification sample and the action taken by the manufacturer with regard to the failure. If the manufacturer does not correct and return to the qualifying activity inverters rejected during qualification or qualification retention testing within twenty-one (21) calendar days after the manufacturer receives them or does not submit a program acceptable to the qualifying activity for the correction of deficiencies noted within twenty-one (21) calendar days, the manufacturer will be notified of the proposed cancellation of the qualification program or proposed removal of the item from the Qualified Products List, as appropriate.

4.6 Quality conformance inspections. Quality conformance inspections shall consist of inspection of the product for delivery. Each inverter offered for deli y shall be subjected to the inspection specified in Table III. Data from t e inspections shall be recorded and be made available for examination by authorized Government representatives.

4.6.1 Failure. Any inverter failing to comply with specified requirements during the quality conformance inspections shall not be delivered on the contract or purchase order. The manufacturer may take corrective action on the failed inverters as appropriate and resubmit for inspection. Reworked units shall be suitably marked to identify them as repaired items. In the event of failure during reinspection, further rework and resubmission shall not be allowed.

4.7 Methods of inspection.

4.7.1 Test conditions. Unless otherwise specified, tests shall be made under the following conditions.

4.7.1.1 Ambient temperature. Tests shall be performed at an ambient temperature of 25 ± 15°C.

4.7.1.2 Ambient altitude. Tests shall be run at approximately sea level altitude.

4.7.1.3 Tolerance and accuracy. The tolerance of test conditions, other than room ambient, and the accuracy of test apparatus shall be as specified in the test facilities requirements of MIL-E-81910.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

4.7.1.4 Ambient temperature measurement. When tests are required to be conducted at reduced or elevated temperatures, the ambient temperature shall be determined by measurements of the air at several points surrounding, but not above, the inverter, and through averaging the results. For units without fans, air temperature measurements shall be taken 1 ± 1/4 inch from the surface of the inverter. For units with an internal fan, air temperature measurements taken 1 ± 1/4 inch from the fan inlet shall be considered the ambient temperature.

4.7.1.5 Airflow. For all maximum temperature-altitude tests below 20,000 feet conducted on inverters without integral cooling fans, the airflow velocity over the inverter within the test chamber shall be adjusted to approximate natural convection conditions and shall not exceed 100 feet/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance. For tests above 20,000 feet altitude, the airflow may exceed 100 feet/minute, but shall be adjusted to the minimum velocity consistent with maintaining the required ambient temperature within specified tolerance.

4.7.1.6 Stabilization. Stabilization of test conditions based on constant input voltage and ambient temperature is defined to have occurred when the change over a ten (10) minute period is measured to be less than:

a. Output voltage: 0.04 Volts RMS

b. Frequency: 0.1 Hz

c. Inverter critical component temperature: 0.5°C

4.7.1.7 Voltage measurement. The input and output voltage shall be measured at the terminals of the unit.

4.7.1.8 Position. The units shall be mounted base down.

4.7.1.9 Loading. Prior to testing, the inverter shall be operated at full rated load (see 6.4.3) until stabilized. The test shall be performed with the inverter operating at full rated load.

4.7.1.9.1 Load power factor. Unity power factor load shall be applied.

4.7.1.9.2 Multiple phase loading. For three-phase tests, the loads on each of the three phases shall be balanced with each phase current within 3 percent of the average magnitude of all three phase currents.

4.7.1.9.3 Definition of load. Single-phase load shall be considered the product of the RMS line current and the RMS voltage measured at the terminals of the inverter. Three-phase load shall be the $\sqrt{3}$ times the product of the average RMS line current and the average RMS line-to-line voltage measured at the terminals of the inverter.

4.7.1.9.4 Definition of full load. Full load shall be the maximum continuous load specified at the applicable input voltage and temperature-altitude conditions specified in the applicable specification sheet.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:49Z
Check the source to verify that this is the currentversion before use.

4.7.1.10 Input voltage. Nominal rated input voltage shall be applied (see 3.7.1.1).

4.7.2 Methods of test.

4.7.2.1 Visual and mechanical inspection. The inverter shall be inspected to determine compliance with the requirements of this specification and the applicable specification sheet with respect to materials, design and construction, parts, workmanship, dimensions, weight, and markings for which there are no applicable tests.

4.7.2.2 Voltage and frequency regulation. The output voltage and frequency shall be measured under the conditions specified in 4.7.2.2.1 through 4.7.2.2.6. For each load, input voltage, position, and temperature-altitude condition, output voltage and frequency shall remain within the limits specified (see 3.7.1.2 and 3.7.1.3). During the course of the tests, no components of the inverter shall exceed their maximum rated temperatures. Tests may be conducted at any intermediate input voltage, load, or temperature-altitude condition within the range of MS33543 Curve I or II (as applicable) at the option of the activity responsible for qualification.

4.7.2.2.1 Voltage and frequency regulation at room ambient temperature. The following tests, (a) through (d) below, shall be conducted immediately upon energizing the inverter, and then repeated after operating the inverter at nominal input voltage, full load, unity power factor until the output voltage and frequency are completely stabilized.

a. At the maximum normal input voltage (see 3.7.1.1a), the load shall varied from no load to rated full load in steps of one-fourth rating. At each lo condition, the inverter shall demonstrate its ability to activate.

b. At nominal rated input voltage (see 3.7.1.1), the test specified in 4.7.2.2.1a shall be repeated.

c. At the minimum normal input voltage (see 3.7.1.1a), the test specified in 4.7.2.2.1a shall be repeated.

d. At the minimum abnormal input voltage (see 3.7.1.1b), the inverter output load current shall be varied from no load to rated full load current (computed on the basis of 115 Volts RMS output) in one-fourth load increments. At each load condition, the inverter shall demonstrate its ability to activate.

4.7.2.2.2 Power factors. The tests specified in 4.7.2.2.1 shall be repeated at the minimum lagging and minimum leading power factors specified in the applicable specification sheet. Tests will be made at any other load power factors between minimum lagging and minimum leading at the option of the activity responsible for qualification.

4.7.2.2.3 Voltage and frequency regulation at maximum temperature-altitude conditions. The inverter shall be tested under each of the temperature altitude conditions listed in Table IV. Prior to being energized, the inverter shall be soaked at the initial maximum temperature, sea level condition for at least one-half hour. Immediately after energizing the inverter, the test specified in 4.7.2.2.1 shall be conducted. The tests of 4.7.2.2.1 and 4.7.2.2.2 shall then be conducted after operating

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
Check the source to verify that this is the current version before use.

peak deviation of 5 KHz at full rated audio output. Wideband FM produces a

the inverter at nominal input voltage, full load, unity power factor until inverter output voltage, frequency, and temperature are completely stabilized. For each subsequent temperature-altitude condition, the test of 4.7.2.2.1 need only be conducted after inverter stabilization.

4.7.2.2.4 Voltage and frequency regulation at minimum temperature-altitude conditions. The tests specified in 4.7.2.2.3 shall be repeated substituting the temperature-altitude conditions of Table V and increasing the soak time at the initial -55°C sea level condition to two hours.

4.7.2.2.5 Operating position. The inverter shall be operated at nominal input voltage, full load, unity power factor in the normal base down position until output voltage, frequency, and critical component temperatures stabilize. Output voltage, frequency, and critical component temperature shall be measured as the inverter is rotated by increments of 90 degrees from the normal base down position, 360 degrees about each major axis except the vertical axis. Each position shall be held until output voltage, frequency, and critical component temperatures stabilize.

4.7.2.2.6 Worst condition. The inverter shall be operated at the worst indicated conditions determined from the tests specified in 4.7.2.2.1 through 4.7.2.2.5 over the normal input voltage range. The conditions from each of these tests resulting in highest output voltage shall be simultaneously applied. Prior to the test, the inverter shall be soaked at the applicable ambient temperature for a minimum of two hours prior to energizing the inverter. The inverter shall be operated until voltage, frequency, and critical component temperatures stabilize. The above test shall be repeated for minimum voltage, maximum frequency, minimum frequency, and maximum critical component temperature conditions.

4.7.2.3 Efficiency. The power output and input of the unit shall be measured at full load, unity power factor, sea level, and nominal input voltage. The efficiency shall be calculated under these conditions and shall not be less than specified in 3.7.1.4 under stabilized test conditions as defined in 4.7.1.6. When computing efficiency, mean (average) values of input voltage and current shall be used, and root mean square (RMS) values of the output voltage and current shall be used.

4.7.2.4 Three-phase output. (three-phase inverters only).

4.7.2.4.1 Output voltage unbalance.

4.7.2.4.1.1 Definition. Output voltage unbalance is defined as the maximum difference between phase voltage magnitudes.

4.7.2.4.1.2 Unbalanced loads. The effects of single-phase and unbalanced three-phase loads on the balance of the three-phase voltage output of the unit shall be determined as follows at minimum, nominal, and maximum normal input voltage.

a. When the unit is carrying no three-phase load, a single-phase line-to-neutral unity power factor load requiring 1/3 and then 2/3 of rated full load output current (a load of approximately 1/9 and 2/9 of the three-phase rating of the unit) shall be connected successively to each phase.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

b. When the unit is carrying a unity power factor balanced three-phase load requiring 1/3 of the rated full load output current, a single-phase line-to-neutral unity power factor load requiring 1/3 and then 2/3 of the rated full load output current (a load of approximately 1/9 and 2/9 of the full three-phase rating of the unit) shall be connected successively to each phase.

c. When the unit is carrying a unity power factor balanced three-phase load requiring 2/3 of the rated full load output current, a single-phase line-to-neutral unity power factor load requiring 1/3 of the rated full load output current (a load of approximately 1/9 of the full three-phase rating of the unit) shall be connected successively to each phase.

The maximum value of the output voltage unbalance and the phase voltage occurring during parts (a) through (c) above shall not exceed the limits specified in 3.7.1.5.2.

4.7.2.4.1.3 Balanced loads. For all balanced loads within the power factor and load rating of the unit, the three-phase output voltage shall comply with 3.7.1.2 and 3.7.1.5.2. This shall be demonstrated during the test specified in 4.7.2.2.

4.7.2.4.2 Phase rotation. In conjunction with the test specified in 4.7.2.4.1, the phase rotation shall be determined and verified for compliance with 3.7.1.5.1.

4.7.2.4.3 Phase displacement angle. In conjunction with the tests specified in 4.7.2.2.1, 4.7.2.2.2, and 4.7.2.4.1.2, the phase displacement angle shall be determined and verified for compliance with 3.7.1.5.3.

4.7.2.5 Recovery time. The output voltage transient shall be measured when rated full load is applied and removed at unity and minimum lagging power factors. The unit shall not develop sustained oscillations and the output voltage shall remain within the limits specified in 3.7.1.6. This test shall be conducted at minimum, ambient, and maximum temperatures specified at approximately sea level pressure and minimum, nominal, and maximum normal input voltages (see 3.7.1.1a).

4.7.2.6 Input power application.

4.7.2.6.1 Minimum voltage. By the addition of external resistance in the line, the voltage input to the inverter from a regulated 28 VDC source shall be reduced to the minimum abnormal input voltage (see 3.7.1.1b). The inverter shall demonstrate the ability to activate and operate continuously with this external resistance at zero, half, and full load.

4.7.2.6.2 Input power application current and voltage. The input power application current and output voltage shall be measured at minimum, nominal, and maximum normal input voltage (see 3.7.1.1a) while initially applying power to the unit at room ambient temperature at zero, half, and full rated load at unity and minimum lagging power factors. The input power application current and voltage measurements shall indicate compliance with the limits specified in 3.7.1.7. This test shall be conducted at minimum, maximum, and room ambient temperatures specified at approximately sea level pressure.

4.7.2.7 Overload and short circuit capacity. The inverter shall be tested to demonstrate compliance with the overload and short circuit capacity requirement-

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

specified in 3.7.1.8. Overload tests shall be conducted at both unity and minimum lagging power factors. The five second (or 15 second as applicable) overload or short circuit test shall be conducted at maximum, minimum, and ambient temperatures specified at approximately sea level pressure. Compliance to the 15 minute duration overload or short circuit requirement of 3.7.1.8.3 shall be demonstrated at minimum and room ambient temperatures only.

4.7.2.8 Electromagnetic compatibility. Conducted and radiated electromagnetic interference shall be measured in accordance with the conducted interference test of MIL-E-81910. During this test, the inverter load power factor shall be adjusted such that the combined power factor of the load and all auxiliary test equipment required shall present a total load of unity power factor to the inverter. The test shall be conducted at full and half load, at minimum and maximum normal input voltage.

4.7.2.9 Output modulation. The unit shall be operated at the conditions specified in 4.7.2.2.1 through 4.7.2.2.4 and shall meet the requirements of 3.7.1.10, except that frequency modulation need only be measured during the tests of 4.7.2.2.1 and 4.7.2.2.2.

4.7.2.9.1 Voltage modulation. Total output voltage modulation shall be determined by means of a calibrated oscilloscope or other means satisfactory to the responsible qualifying activity. The percent modulation shall be computed as follows:

$$\text{Percent modulation} = \frac{\text{Emax} - \text{Emin}}{\text{Emax} + \text{Emin}} \times 100\%$$

Emax and Emin are the maximum and minimum peak voltages respectively, measured for the same polarity during a one second time period.

4.7.2.9.2 Frequency modulation. Output frequency modulation shall be determined by a method acceptable to the responsible qualifying activity. The percent modulation shall be computed as follows:

$$\text{Percent modulation} = \frac{\text{Fmax} - \text{Fmin}}{\text{Fmax} + \text{Fmin}} \times 100\%$$

Fmax and Fmin are the maximum and minimum instantaneous frequencies, respectively, attained during a 15 second time period.

4.7.2.10 Waveform.

4.7.2.10.1 Distortion. The harmonic and extraneous frequency content of the output voltage, per phase for three-phase units and across the two output lines for single-phase units, shall be determined under the conditions specified in 4.7.2.10.1.1 and 4.7.2.10.1.2. Compliance to the requirements of 3.7.1.11.1 and 3.7.1.11.2 shall be determined.

4.7.2.10.1.1 Linear loads (full wave sinusoidal output current).

a. Under the test conditions specified in 4.7.2.2.1a,b,and c with unity power factor, steady-state load.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
Check the source to verify that this is the current version before use.

b. Under the test conditions specified in 4.7.2.2.1a, b, and c with steady-state minimum lagging power factor load specified in the applicable specification sheet.

c. Tests (a) and (b) above, shall be repeated at maximum and minimum rated temperatures at approximately sea level pressure.

4.7.2.10.1.2 Non-linear loads. The tests specified below shall be conducted at minimum, nominal, and maximum normal input voltages (see 3.7.1.1a). Tests on three-phase units may be conducted one phase at a time.

a. With an initial linear load of 50 percent rated current, an additional load of 50 percent rated current shall be added for approximately 1/2 of each positive and negative half cycle.

b. Part (a) shall be repeated except that the load during the positive half cycle shall be linear and 100 percent rated current.

c. With no initial load, a load of 50 percent rated current shall be applied for approximately 1/2 of each positive and negative half cycle.

d. Part (c) shall be repeated except that the load during the positive half shall be linear and 50 percent rated current.

4.7.2.10.2 Crest factor. The crest factor shall be measured under the test conditions specified in 4.7.2.2.1 and 4.7.2.2.2 to verify compliance with 3.7.1.11.1 The crest factor limits apply to either polarity separately.

4.7.2.11 Input ripple current. The inverter shall be tested to verify compliance with 3.7.1.12 at zero, half, and full load at minimum, nominal, and maximum normal input voltages.

4.7.2.12 Input power susceptibility.

4.7.2.12.1 Transient spike susceptibility. The inverter shall be operated at no load and full load at 28 VDC input. Spike voltages with repetition rates of 1, 10, 25, and 50 pulses per second shall be applied to the inverter input for both positive and negative polarities. The test duration shall be one minute for each combination of loading, pulse repetition rate, and spike polarity. The spike generator shall have a source impedance of 50 ±5 Ohms, and shall produce a waveform as shown in Figure 2 when operating into an open circuit prior to connection to the inverter input. The spike waveform at the inverter terminals during the test shall be recorded. During the test the inverter shall be observed for evidence of flashover or insulation breakdown, and the output shall be monitored to verify compliance with 3.7.1.13. Subsequent to the test, the output voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c.

4.7.2.12.2 Transient surge susceptibility. With the inverter operating at no load and full load with an input of 28 VDC, a positive step voltage of 80 Volts with a duration of 50 milliseconds shall be applied to the inverter input. The transient surge shall be obtained from a low impedance source, such as a capacitor bank. The transient surge shall be applied four consecutive times for each load condition. The magnitude

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
Check the source to verify that this is the current version before use.

and duration shall be monitored on an oscilloscope and shall be measured, prior to application to the inverter under test, by applying and adjusting the transient across a dummy load which approximates the input of the inverter under test. The inverter shall be observed for evidence of flashover or insulation breakdown, and the output shall be monitored to verify compliance with 3.7.1.13. The inverter output voltage shall be permitted to drop to any level during the duration of the surge. Subsequent to the test the output voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c.

4.7.2.12.3 Ripple voltage susceptibility. With the inverter operating at no load, 1/4, 1/2, 3/4, and rated full load, unity power factor, at minimum, nominal, and maximum normal input voltage (see 3.7.1.1a), an audio frequency signal shall be superimposed on the inverter input and shall be varied from 10 Hertz to 30 KHertz. The amplitude of the signal shall conform to Figure 3. At the option of the qualifying activity the inverter may be operated from any power source whose individual frequency characteristics conform to Figure 3, and whose total peak-to-mean ripple is 2.0 Volts or less. The output of the inverter shall be monitored throughout the test in order to measure output voltage, frequency, modulation, and waveform to verify compliance with 3.7.1.2, 3.7.1.3, 3.7.1.10, and 3.7.1.11.1. The test shall then be repeated at minimum rated lagging power factor load.

4.7.2.13 Extreme input voltage. With the inverter operating at no load, the input shall be varied between zero and 32 Volts over a five minute period. This test shall be repeated with the inverter operating at rated full load, unity power factor. The inverter shall then be operated for 1/2 hour at no load at each of the following input voltages: 32, 15, 10, and 5 VDC. This test shall be repeated with the inverter operating at rated full load, unity power factor. The output shall be monitored throughout the test to verify compliance with 3.7.1.14.

4.7.2.14 Acoustical noise. The acoustical noise of the inverter shall be measured in planes one foot from each of the six sides. In each plane, measurements shall be made opposite the center of the inverter surface to determine the sound level. The maximum sound level at all points shall be measured using the B weighted scale of an integrating-type sound level meter. Microphone orientation shall be in accordance with the recommendations of the manufacturer of the sound level meter to obtain the required frequency response. The calibrating devices or standards used shall be directly traceable to the National Institute of Science and Technology. Measurements shall be made under anechoic conditions with the inverter supported in a manner to minimize acoustic transmission and reflection by the support structure. Compliance to 3.7.1.15 shall be determined.

4.7.2.15 Acceleration. The inverter shall be tested in accordance with the acceleration test of MIL-E-81910. The inverter shall be operated at 28 VDC input and full load during Procedure II, and the output voltage and frequency shall be monitored during the test to determine compliance with 3.7.2.2. At the conclusion of the test, the inverter shall be examined for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c.

4.7.2.16 Humidity. The inverter shall be tested in accordance with the humidity test of MIL-E-81910. Following the test, the voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c. The inverter shall be inspected for evidence of damage.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

4.7.2.17. Shock. The inverter shall be tested in accordance with the mechanical shock test of MIL-E-81910, except that the inverter shall be operating during the basic design test at full load and 28 VDC input. The output voltage shall be monitored during the test to determine compliance with 3.7.2.4.1. At the conclusion of the test, the inverter shall be inspected for evidence of damage, and voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c. The inverter shall not be operated during the crash safety test.

4.7.2.18 Vibration.

4.7.2.18.1 Inverters weighing less than 35 lbs. The inverter shall be subjected to sinusoidal vibration applied to the mounting base. The amplitude shall be .015 inch (maximum total excursion .03 inch) up to 60 Hertz. Above 60 Hertz the acceleration shall be 10g. The inverter shall be vibrated for one excursion over the frequency range of 20 to 2000 Hertz along each of its major axes. The vibration excursion shall take 15 minutes for each axis. Dwell time at resonance (20 to 2000 Hertz) shall be 3 minutes at the worst resonance condition detected in each axis. During the test, the inverter shall be operated at rated full load and 28 VDC input. The output voltage modulation shall not exceed two percent, and output voltage and frequency shall not be outside the limits of 3.7.1.2 and 3.7.1.3 at any time during the test. Following the test the inverter shall pass the test specified in 4.7.2.2.1a, b, and c, and shall be thoroughly inspected for cracking or rupture of any components, solder connections, or of the mounting assembly itself. The presence of such damage shall be cause for rejection.

4.7.2.18.2 Inverters weighing 35 lbs or more. The inverter shall be vibrated in accordance with MIL-STD-810, Method 514.2, Procedure I. Procedure I shall be accomplished using vibration test Curve C of Figure 514.2-2 and the time schedule Table 514.2-II for the frequency range of 5 to 2000 Hertz. The inverter shall be mounted without vibration isolators. During the test, the inverter shall be operated at rated full load and 28 VDC input. The output voltage modulation shall not exceed two percent, and output voltage and frequency shall not be outside the limits of 3.7.1.2 and 3.7.1.3 at any time during the test. Following the test, the inverter shall pass the test specified in 4.7.2.2.1a, b, and c, and shall be thoroughly inspected for cracking or rupture of any components, solder connections, or of the mounting assembly itself. The presence of such damage shall be cause for rejection.

4.7.2.19 Fungus. The inverter and installation instructions shall be tested in accordance with the fungus test of MIL-E-81910. At the conclusion of the test, the inverter shall be inspected for fungus growth, and the output voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c.

4.7.2.20 Salt fog. The inverter shall be tested in accordance with the salt fog test of MIL-E-81910. Specified drying shall be allowing the unit to stand at room ambient conditions for 24 hours. Output voltage and frequency regulation shall then be measured as specified in 4.7.2.2.1a, b, and c. This regulation test shall be repeated after 48 hours additional exposure at room ambient conditions.

4.7.2.21 Dust. The inverter shall be tested in accordance with the dust test of MIL-E-81910. The inverter shall be operated at no load with an input voltage of 28 VDC, when required. At the conclusion of the test, the output voltage and frequency regulation shall be measured as specified in 4.7.2.2.1a, b, and c. For units equipped with a cooling fan, proper operation (speed and noise) shall be verified.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

4.7.2.22 Explosive atmosphere. The inverter shall be tested in accordance with MIL-STD-810, Method 511.3, Procedure I.

4.7.3 Reliability development. The reliability development test shall consist of the tests specified in 4.7.3.1 and 4.7.3.2. The test shall be structured as a test, analyze and fix (TAAF) program wherein any failure (any operation outside of specified limits) which occurs is analyzed and corrective action implemented by the manufacturer to prevent recurrence of the failure. A minimum of 4000 hours total inverter operating time on two samples is required with two or less failures in order to successfully complete the test. In the event more than two failure occur during the 4000 hour period, additional inverter operation under the conditions of 4.7.3.2 is required until the total inverter operating hours divided by the number of failures (cumulative MTBF) equals 2000 or higher. When a corrective action has resolved a failure to the satisfaction of the qualifying activity all but the first occurrence of the resolved failure shall be dropped from consideration. No single inverter of the two samples shall accumulate less than 25 percent of the total test time, and both inverters shall complete the entire test of 4.7.3.1.

4.7.3.1 Temperature cycling. Two sample inverters shall be operated for 50 hours under the input voltage, load, and duty cycle profile shown in Figure 4. Twenty-five of the operating hours shall be at unity power factor, and the remaining 25 hours at the minimum lagging power factor shown on the applicable specification sheet. During the test the ambient temperature shall be varied continuously between -55°C and +71°C with the extreme temperatures maintained for 1 ± .25 hours prior to ramping to the opposite extreme temperature. The heating and cooling ramps shall be performed at a minimum rate of 3°C/minute. This test may be performed continuously or in increments equal to one complete temperature cycle. During the test, the output voltage and frequency shall be monitored for compliance with 3.7.1.2 and 3.7.1.3. Following the test, the inverter shall pass the test of 4.7.2.2.1.

4.7.3.2 Room ambient operation. The two sample inverters shall be operated at room ambient conditions under the input voltage, load, and duty cycle profile shown in Figure 4 for the time duration required by 4.7.3. During the test, the output voltage and frequency shall be monitored for compliance with 3.7.1.2 and 3.7.1.3. Each inverter shall be subjected to the test of 4.7.2.2.1 after each 250 operating hours. Following the entire test duration, each inverter shall pass the following tests at room temperature only: 4.7.2.2.1, 4.7.2.2.2, 4.7.2.3, 4.7.2.7, 4.7.2.9, 4.7.2.10.1.1a and b.

4.7.4 Disassembly and inspection. At the conclusion of the test program, each sample shall be disassembled, as necessary, to inspect for evidence of wear, defects, and overheating. The presence of such evidence shall be cause for rejection.

4.7.5 Burn-in. The inverter shall be tested in accordance with the burn-in test of MIL-E-81910.

5. PACKAGING.

5.1 Preservation and packaging. The inverter shall be packaged in accordance with MIL-STD-2073-1, level A or C as specified in the contract order.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
Check the source to verify that this is the current version before use.

5.1.1 Level A. For level A packaging, preservation shall be in accordance with MIL-P-116, Method II, without preservation compounds, using metal reusable containers unless otherwise specified.

5.1.2 Level C. For level C the inverter shall be preserved and packaged individually in accordance with the manufacturer's commercial practice.

5.2 Packing. Packing shall be in accordance with MIL-STD-2073-1 ,level A, B,or C as specified in the contract or order.

5.3 Marking. Interior and exterior containers shall be marked in accordance with MIL-STD-2073-2 and MIL-STD-129.

5.3.1 Reinspection dates. Reinspection date marking shall be as specified by the procuring activity.

6. NOTES AND CONCLUDING MATERIAL.

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The items described herein are intended for use on military aircraft to convert 28 Volt DC power to either single or three-phase, 115 VRMS, 400 Hertz AC power. Inverters typically are used to provide AC power on aircraft with main electrical generating systems that provide DC power.

6.2 Acquisition requirements.

a. Title, number, and date of this specification.

b. Class.

c. Military part number.

d. Required level of packaging and packing.

e. Type of container.

f. Reinspection date marking and any special marking.

g. Handbooks or manuals, when required.

h. That the inverter shall have a minimum time to first maintenance action of 500 flight hours.

i. That the manufacturer shall include the cost of units required for retention of qualification (see 4.5) in the amount of his bid.

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time set for the opening of bids, qualified for inclusion in the applicable Qualified Products List - whether or not such products have actually been so listed by that date. The attention of the manufacturers is cal?

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:49Z
Check the source to verify that this is the currentversion before use.

to these requirements, and manufacturers are urged to arrange to have the products tha they propose to offer to the Federal Government tested for qualification in order tha they may be eligible to be awarded contracts or orders for the products covered by thi specification. The activity responsible for the Qualified Products List is the Nava Air Systems Command (AIR-5363), Department of the Navy, Washington, D.C. 20361-5360 however, information pertaining to qualification of products may be obtained from th qualifying activity - the Naval Air Warfare Center Aircraft Division, Electrica Systems Department (SY60), Patuxent River, MD 20670-5304.

6.3.1 Qualification retention samples. For each quotation and based on al production contracts obtained, it will be the manufacturer's responsibility to infor the procuring activity of the exact number of qualification retention samples require for forwarding to the qualifying activity (see 4.5). The number of qualificatio retention samples required should be taken into account when the contract bid i prepared. If the proper number of retention samples is not itemized in the contract then the manufacturer will, without additional cost to the Government, forward to th qualifying activity whatever number of retention samples is determined by the cognizan Government Inspector to be required by paragraph 4.5 of this specification.

6.4 Definitions.

6.4.1 Alternating voltage and current. Alternating voltages and currents used i this specification shall be root mean square (RMS) values, unless otherwise specified

6.4.2 Direct voltage and current. Direct voltages and currents used in thi specification shall be mean (average) values.

6.4.3 Full rated load (full load). Full rated load used in this specificatio shall be the maximum continuous inverter output load current specified in th applicable specification sheet for a given temperature-altitude condition.

6.4.4 Phase voltage. Phase voltage shall be considered to be line-to-neutral fo wye connection and line-to-line for delta connection.

6.5 Installation. The aircraft manufacturer shall allow clearance of at least on inch at each end of the inverter in addition to the maximum dimensions shown on th applicable specification sheet for installation and ventilation.

6.6 Three-phase application. For three-phase aircraft application, the load o any one phase shall not exceed one third of the total three-phase inverter rating o a steady-state basis.

6.7 Subject term (key word) listing.

AC supply
Electrical
Single-phase
Static supply
Three-phase

6.8 Changes from previous issue. Marginal notations are not used in this revisio to identify changes with respect to the previous issue due to the extensiveness of th changes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

Table I. Component derating.

Part Type	Critical Stress	Maximum Allowable Application Stress
MICROCIRCUITS		
Digital	Output Current	.80
	Operating Frequency	.75
Linear	Bias Voltage	.70
	Input Signal Voltage	.70
	Output Current	.75
	Operating Frequency	.75
Voltage Regulator	Input Voltage (Rated Maximum)	.80
	Output Current (Rated Maximum)	.75
	Power Dissipation (Rated Maximum)	.60
TRANSISTORS		
General Purpose	Power	.50
	Current	.75
	Voltage	.75
Power	Power	.30
	Current	.75
	Voltage	.75
Switching	Power	.50
	Current	.75
	Voltage	.75
DIODES		
General Purpose	Power	.50
	PIV	.50
	Surge Current	.50
	FWD Current	.50
Rectifier	Power	.30
	PIV	.50
	Surge Current	.50
	FWD Current	.50

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

Table I. Component derating (Continued)

Part Type	Critical Stress	Maximum Allowable Application Stress
DIODES		
Switching	Power	.30
	PIV	.50
	Surge Current	.50
	FWD Current	.50
SCR	Power	.30
	PIV	.50
	Surge Current	.50
	FWD Current	.50
Varactor	Power	.50
	PIV	.75
	FWD Current	.75
Zener	Power	.50
	FWD Current	.50
	Zener Current	.5 [I_z max + I_z nom]
Reference	Power	.30
	FWD Current	.50
	Zener Current	.5 [I_z max + I_z nom]
CAPACITORS, FIXED		
Ceramic	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Mica, Dipped	Voltage	.60
	Current	
	AC/Ripple	.70
	Surge	.70
Glass	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

TABLE I. Component derating (Continued).

Part Type	Critical Stress	Maximum Allowable Application Stress
CAPACITORS, FIXED		
Porcelain	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Paper	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Plastic	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Tantalum, Foil or Film	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Tantalum, Solid, Dry	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
	Protection	3 OHMS/VOLT
Aluminum Electrolytic	Voltage	.65
	Current	
	AC/Ripple	.70
	Surge	.70

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

TABLE I. Component derating (Continued).

Part Type	Critical Stress	Maximum Allowable Application Stress
CAPACITORS, ADJUSTABLE		
Ceramic	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Glass	Voltage	.50
	Current	
	AC/Ripple	.70
	Surge	.70
Air	Voltage	.30
RESISTORS, FIXED		
Carbon Composition	Power	.50
Insulated Film	Power	.50
Wirewound Precision		
1.0% Tol.	Power	.50
.1% Tol.	Power	.25
Wirewound Power	Power	.50
Thermistor	Power	.50
RESISTORS, ADJUSTABLE		
Wirewound	Rated Current	.70
Non Wirewound	Rated Current	.70

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

TABLE I. Component derating (Continued).

Part Type	Critical Stress	Maximum Allowable Application Stress
RESISTORS, ALL	Voltage	.80
MISCELLANEOUS		
Transformers and Chokes	Flux Saturation level	.75
	Insul. Breakdown Volt.	.50
	Current	.70

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

TABLE II. Qualification inspection.

INSPECTION	INVERTER SAMPLE				REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
	1	2	3	4		
Visual & mechanical inspection	X	X	X	X	3.4, 3.5, 3.6, 3.9, 3.10	4.7.2.1
Voltage and frequency regulation					3.7.1.2, 3.7.1.3	4.7.2.2
Room ambient temp-alt	X*	X*	X	X		4.7.2.2.1
Power factors	X*	X*				4.7.2.2.2
Maximum temp-alt	X*	X*				4.7.2.2.3
Minimum temp-alt	X*	X*				4.7.2.2.4
Operating position	X	X				4.7.2.2.5
Worst condition	X	X				4.7.2.2.6
Efficiency	X*	X*	X	X	3.7.1.4	4.7.2.3
Three-phase output	X*	X*	X	X	3.7.1.5	4.7.2.4
Recovery time	X	X	X	X	3.7.1.6	4.7.2.5
Input power application		X		X	3.7.1.7	4.7.2.6
Overload & short circuit capacity	X*	X*	X	X	3.7.1.8	4.7.2.7
Electromagnetic compatibility	X		X		3.7.1.9	4.7.2.8
Output modulation	X*	X*	X	X	3.7.1.10	4.7.2.9
Waveform	X*	X*	X	X	3.7.1.11	4.7.2.10
Input ripple current	X	X	X	X	3.7.1.12	4.7.2.11
Input power susceptibility	X	X			3.7.1.13	4.7.2.12
Extreme input voltage	X	X			3.7.1.14	4.7.2.13
Acoustical noise	X				3.7.1.15	4.7.2.14
Acceleration	X				3.7.2.2	4.7.2.15
Humidity	X				3.7.2.3	4.7.2.16
Shock	X				3.7.2.4	4.7.2.17
Vibration		X	X		3.7.2.5	4.7.2.18
Fungus				X	3.7.2.6	4.7.2.19
Salt fog		X			3.7.2.7	4.7.2.20
Dust				X	3.7.2.8	4.7.2.21
Explosive atmosphere (when specified)			X		3.7.2.9	4.7.2.22
Reliability development			X	X	3.8	4.7.2.23
Disassembly & inspection			X	X	---	4.7.2.24

Note: For three-phase units, all tests shall be conducted in the three-phase wye configuration. For units designed to operate in both wye and delta configuration, tests indicated by * shall be conducted in both delta and wye configurations.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

TABLE III. Quality conformance inspection.

INSPECTION	REQUIREMENT PARAGRAPH	TEST METHOD PARAGRAPH
Visual & mechanical inspection	3.4, 3.5, 3.6, 3.9, 3.10	4.7.2.1
Burn-in	3.10.1	4.7.4
Overload & short circuit capacity (room temperature only)	3.7.1.8	4.7.2.7
Voltage and frequency regulation at room ambient temperature-altitude	3.7.1.2, 3.7.1.3	4.7.2.2.1
Power factors	3.7.1.2, 3.7.1.3	4.7.2.2.2
Efficiency	3.7.1.4	4.7.2.3
Phase rotation (three-phase units only)	3.7.1.5.1	4.7.2.4.2
Phase balance (three-phase units only)	3.7.1.5.2	4.7.2.4.1
Output voltage modulation	3.7.1.10.1	4.7.2.9.1

Note: For three-phase units, all tests shall be conducted in the three-phase wye configuration.

Source: https://assist.dla.mil -- Downloaded: 2026/10-26T04:49Z
Check the source to verify that this is the current version before use.

TABLE IV. Maximum Temperature-Altitude Conditions.

CLASS A INVERTERS		CLASS B INVERTERS	
AMBIENT TEMPERATURE °C	ALTITUDE FEET	AMBIENT TEMPERATURE °C	ALTITUDE FEET
+71	Sea level	+85	Sea level
+25	15,000	+85	17,500
+8	25,000	+50	35,000
-10	35,000	+20	50,000
-10	50,000	+20	65,000

TABLE V. Minimum Temperature-Altitude Conditions.

CLASS A AND B INVERTERS

AMBIENT TEMPERATURE °C	ALTITUDE FEET
-55	Sea level
-55	10,000
-55	25,000
-55	65,000

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

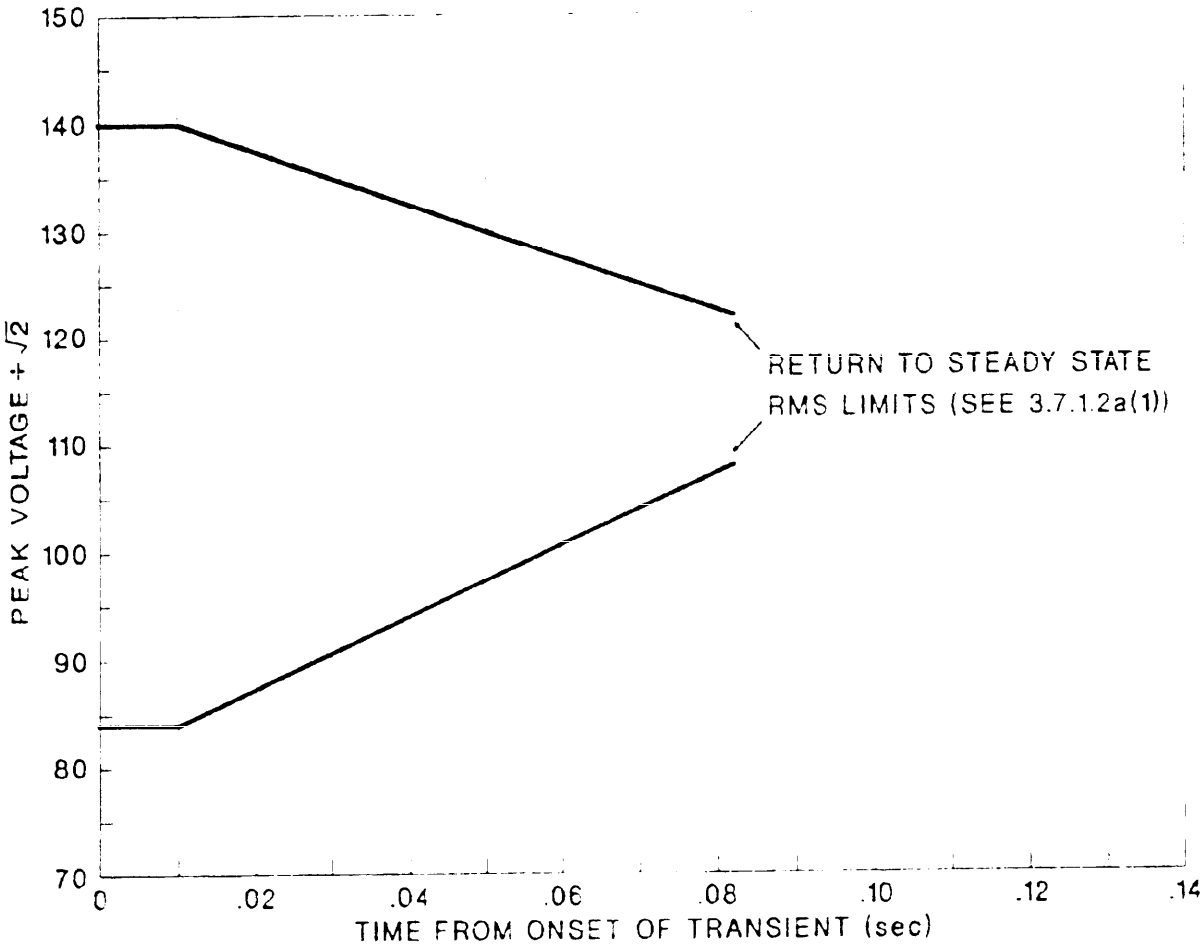


Figure 1. Inverter output voltage transient limits.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

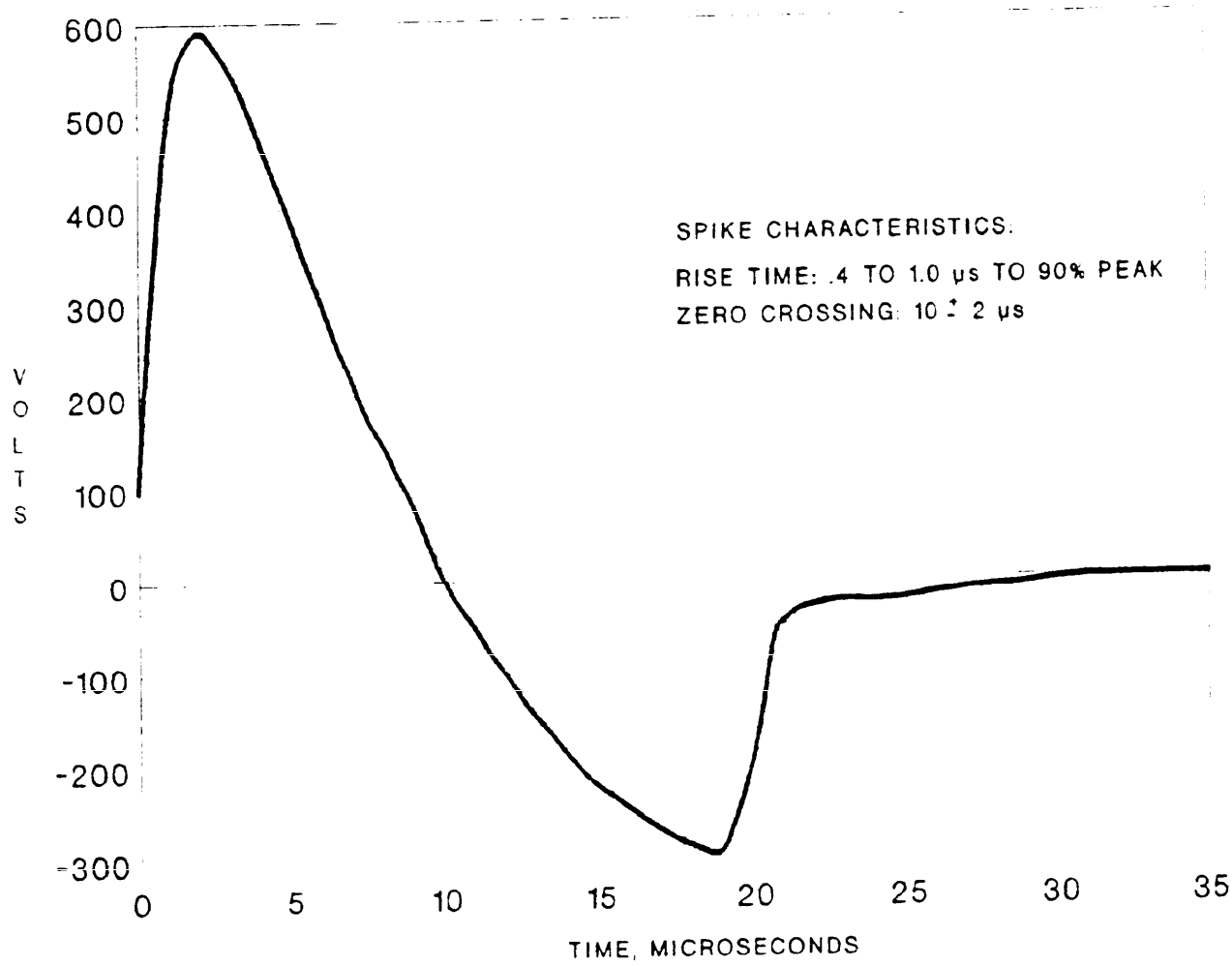


Figure 2. Spike susceptibility test waveform.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

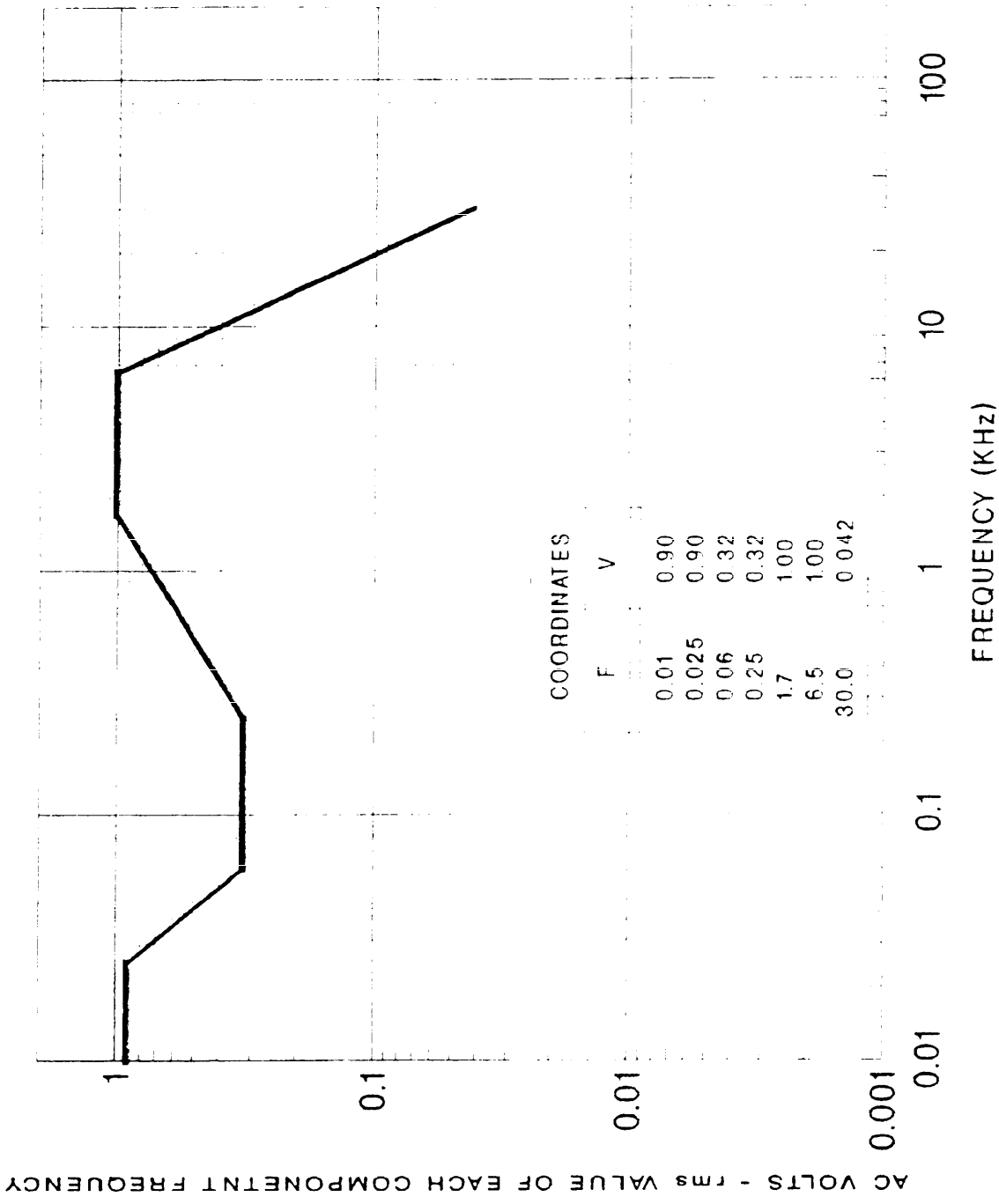


Figure 3. Frequency characteristics of ripple for ripple susceptibility test.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

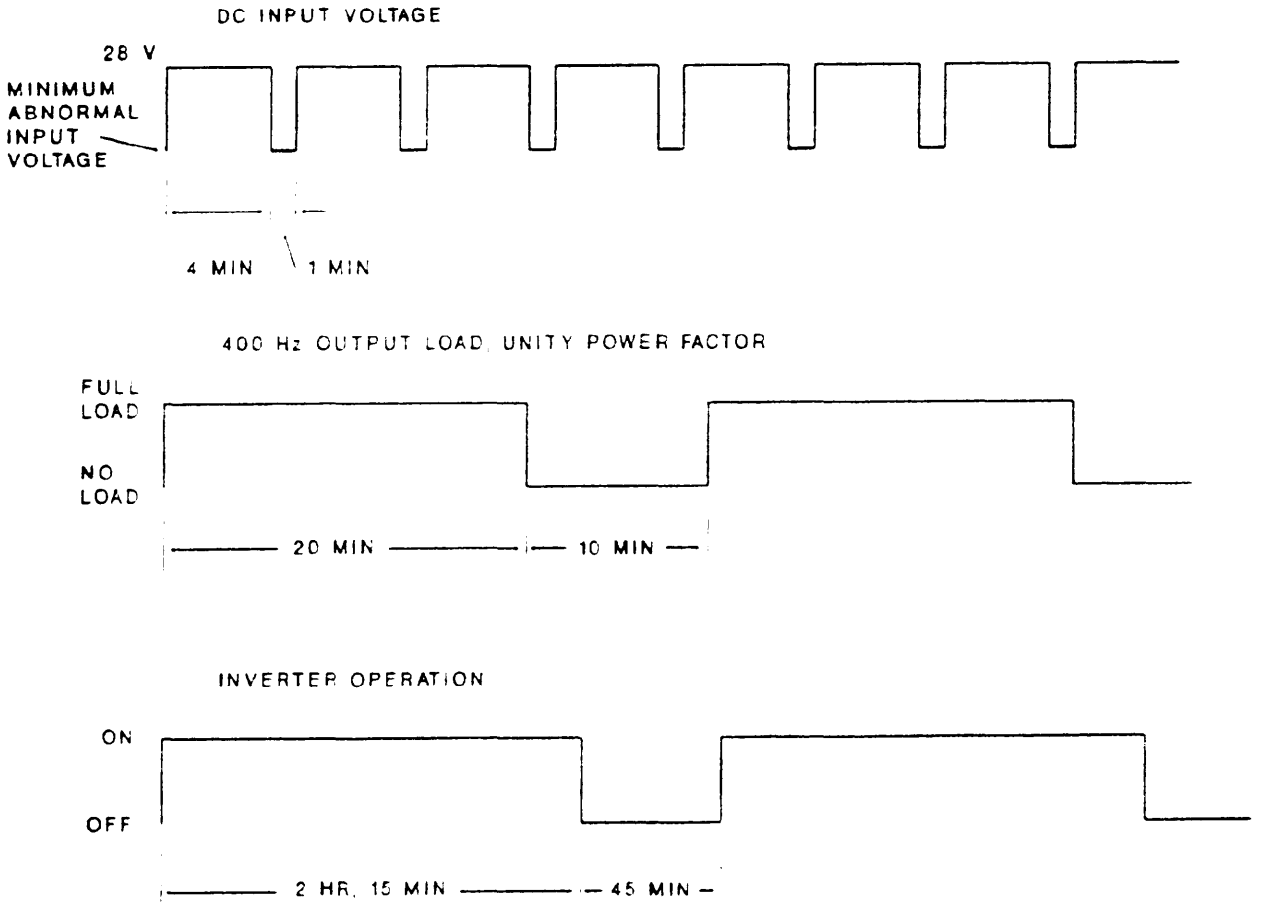


Figure 4. Input voltage, load, and duty cycle profile

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

CONCLUDING MATERIAL

Custodians:
Navy - AS

Preparing Activity
Navy - AS
Project No 6130-N319

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:49Z
Check the source to verify that this is the current version before use.

STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-I-85071A(AS)	920731

3. DOCUMENT TITLE

INVERTERS, AIRCRAFT, DC TO AC, GENERAL SPECIFICATION FOR

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Power Systems (AIR-53632)	(703) 692-2653	222-2653
c. ADDRESS (Include Zip Code) Naval Air Systems Command Jefferson Plaza 2, Room 852 Washington, D.C. 20361-5360	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

DD Form 1426, OCT 89 Previous editions are obsolete. 198

Source: https://assist.dla.mil - Downloaded: 2025-08-26T04:49Z
Check the source to verify that this is the current version before use.